

Roll No.

TBT-701

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

IPR, BIOSAFETY & BIOETHICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Give an overview about traditional knowledge and its relevance in IPR.

(CO1)

OR

(b) What are intellectual property rights ? Explain the types of IPR with examples.

(CO1)

2. (a) 'One of its own kind.' Comment of the aforementioned statement with examples.

(CO1) & (CO3)

OR

(b) What are the benefits covered under Plant Protection Variety Farmer's Act, 2001 ?

(CO1) & (CO3)

P. T. O.

(2)

3. (a) Differentiate between Patent Infringement and Copyright Infringement. (CO2)

OR

- (b) Explain the stepwise process of obtaining a patent in detail. (CO2)
4. (a) "An agreement which covers the trade of international goods." Explain such agreement in detail highlighting its pros and cons also. (CO2)

OR

- (b) What was the role TRIPS agreement ? Describe the process of such agreement. (CO2)
5. (a) Explain the five principles of World Trade Organization. (CO2)

OR

- (b) Discuss few case studies pertaining to the trademark infringement. (CO2)

Roll No.

TBT-702(b)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

RATIONAL DRUG DISCOVERY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain Lipinski's rule and correlate it with pharmacokinetic parameters. (CO1)

OR

- (b) What is virtual screening ? Explain steps of preparation of receptor molecule in Pyrx. (CO1)

2. (a) Write short notes on any *two* chemical databases and their role in drug designing. (CO1)

OR

- (b) Explain major steps of complete drug designing process. (CO1)

P. T. O.

(2)

3. (a) Differentiate between Molecular mechanics and Quantum mechanics.

(CO2)

OR

- (b) Explain various molecular descriptors used in QSAR. What is a pharmacophore ?

(CO2)

4. (a) Describe force fields. Explain types of parameters considered in force field.

(CO2)

OR

- (b) Explain Conjugate variate algorithm for Energy minimization.

(CO2)

5. (a) How is LBDD different from SBDD ? Which approach is more reliable ?

(CO1)

OR

- (b) Explain the following terms :

(CO1)

Clinical trials phases, ADMET, IND, NDE, FDA.

Roll No.

TBT-702(C)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER) MID SEMESTER EXAMINATION, Oct., 2023

ENZYME TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Write a detailed account of enzyme purification from tissue. (CO1)

OR

- (b) Define chromatography. Describe in detail the technique of High-Pressure Liquid Chromatography. (CO1)

2. (a) Explain in detail some of the important characteristics of enzymes. (CO1)

OR

- (b) Define salting out. Describe in detail the technique of column chromatography. (CO1)

3. (a) Explain in details the following terms : (CO1/CO2)
(i) Active site
(ii) Induced Fit Model

P. T. O.

(2)

OR

(b) Derive mathematically the Michaelis-Menten (MM) rate equation.

(CO1/CO2)

4. (a) Describe the mechanism of ES complex formation.

(CO2)

OR

(b) Derive the Lineweaver-Burk (LB) Transformation equation and compare the various parameters of the LB plot and MM plot.

(CO2)

5. (a) Explain in detail the following terms :

(CO2)

(i) Enzyme Kinetics

(ii) Substrate specificity

OR

(b) Explain with the help of a diagram in detail the various features of a substrate velocity plot.

(CO2)

Roll No.

TBT-703(a)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

PHARMACEUTICAL BIOTECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Elaborate on some of the source of drugs. (CO1)

OR

(b) Discuss the different routes of drug administration. (CO1)

2. (a) Define the importance of parenteral route of drug administration. (CO1)

OR

(b) Elaborate the factor affecting route of drug administration. (CO1)

3. (a) What is difference between pharmacokinetics and pharmacodynamics ? (CO1)

OR

(b) Elaborate on the factors that affect the absorption of drug. (CO1)

P. T. O.

(2)

4. (a) What is the meaning of bioavailability ? Explain. (CO1)

OR

- (b) Elaborate on the mechanism of drug action. (CO1)

5. (a) What is role of drug receptor ? Explain. (CO1)

OR

- (b) Write an overview on development of pharmacology. (CO1)

Roll No.

TBT-704

**B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

PYTHON

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What is Python ? Write various features of the Python language. Also explain the potential uses of Python. (CO1/CO3)

OR

- (b) What is a variable ? Write the rules for Python variable names. Explain whether the statement is correct or not : `>>>x, y, z = "Orange", "Banana", "Cherry"`. (CO1/CO3)

2. (a) What is a data type ? List various data types supported by Python. Also explain features of these data types. (CO2/CO4)

OR

- (b) What is a function in Python ? List and explain the use of Python List functions with examples. (CO2/CO4)

P. T. O.

Roll No.

TBT-705(a)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

BIOTERRORISM AND NATIONAL SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What are the differences between ancient and modern terrorism ? (CO1)

OR

(b) What is terrorism ? Define its objectives, goal, and history. (CO1)

2. (a) Explain bioterrorism, its agents, limitations, and regulations for controlling its use. (CO1)

OR

(b) Describe the CBRN weapons with *one* example each. (CO1)

3. (a) Explain the two cases of chemical weapons where mass destruction occurred. (CO1/CO2)

OR

(b) Explain the historic case of nuclear attack by US. (CO1/CO2)

P. T. O.

(2)

4. (a) What is bacteria ? Describe the interaction of bacteria with immune system by taking *one* example. (CO2)

OR

- (b) What is virus ? Describe the interaction of bacteria with immune system by taking *one* example. (CO2)

5. (a) Describe interaction of bioweapon agents with immune system. (CO2)

OR

- (b) How does biological war agents impact the physiology of a human ? Give *one* example. (CO2)

Roll No.

TBT-705(b)

B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

FOOD AND NUTRITION SCIENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Write a short note on Positive interaction of microorganisms with food products. (CO1)

OR

- (b) Illustrate different steps involved in canning process in detail. (CO1)

2. (a) Discuss different intrinsic and extrinsic factor affecting interaction of microorganism with food. (CO1)

OR

- (b) Discuss biochemical and metabolic pathways involved in food production. (CO1)

P. T. O.

lib
10/10/23

4th shift

(2)

3. (a) Discuss the term food **processing** and its various stages with the help of example.

(CO1)

OR

- (b) Write a short note on **malting**, various steps involved and its commercial application.

4. (a) Write name of microorganisms involved in spoilage of meat and poultry products along with their **spoilage** type.

(CO2)

OR

- (b) Illustrate the SCP, its sources, uses and importance.

(CO2)

5. (a) Discuss various steps involved in production of wine.

(CO2)

OR

- (b) Discuss the term probiotics and their mode of action. Also discuss essential features of probiotic microorganisms.

(CO2)